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IT Project Management Support

Lecture 8

Managing Project Risk

AGENDA

- Defininitions
- Project risk management
- Top software development risks
- Risk matrix
- Risk register
- Risk response strategies



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Risk Definitions

Project risk definitions

- There are many definitions of the term "risk" but no generally accepted one.
- All earlier definitions implicitly include two characteristics:
 - **uncertainty** (an event may or may not occur)
 - and **loss** (an event has undesired effects).

This is demonstrated by the following examples:

- **risk** indicates that somebody or something is subject to a danger or certain dangers;
- **risk** is the potential for the occurrence of undesired, negative consequences of an event;
- **risk** is the possibility of suffering losses, damage, disadvantages or destruction;
- **risk** is a future **uncertain** event with a probability of occurrence and potential for **loss**.

Project risk definitions by PMBOK Guide

- Risk exists at two levels within every project.
- Each project contains **individual risks** that can affect the achievement of project objectives.
- It is also important to consider the **riskiness of the overall project**, which arises from the combination of individual project risks and other sources of uncertainty.
- Project Risk Management processes address both levels of risk in projects, and these are defined as follows:
- **Individual project risk** is an uncertain event or condition that, if it occurs, has a **positive or negative** effect on one or more project objectives.
- **Overall project risk** is the effect of uncertainty on the project as a whole, arising from all sources of uncertainty including individual risks, representing the exposure of stakeholders to the implications of variations in project outcome, both **positive** and **negative**.

Project risk definitions by PMBOK Guide

Individual project risks can have a positive or negative effect on project objectives if they occur.

- Project Risk Management aims to exploit or enhance positive risks (opportunities) while avoiding or mitigating negative risks (threats).
- Unmanaged threats may result in issues or problems such as delay, cost overruns, performance shortfall, or loss of reputation.
- Opportunities that are captured can lead to benefits such as reduced time and cost, improved performance, or reputation.

Overall project risk can also be positive or negative.

- Management of overall project risk aims to keep project risk exposure within an acceptable range by reducing drivers of negative variation, promoting drivers of positive variation, and maximizing the probability of achieving overall project objectives.

Project risk management

- Risks will continue to emerge during the lifetime of the project, so Project Risk Management processes should be conducted iteratively.
- Risk is initially addressed during project planning by shaping the project strategy.
- Risk should also be monitored and managed as the project progresses to ensure that the project stays on track and emergent risks are addressed.
- In order to manage risk effectively on a particular project, the project team needs to know what level of risk exposure is acceptable in pursuit of the project objectives.
- This is defined by measurable risk thresholds. Risk thresholds express the degree of acceptable variation around a project objective.
- They are explicitly stated and communicated to the project team and reflected in the definitions of risk impact levels for the project.

Project risk management

- The goal of project risk management can be viewed as minimizing potential negative risks while maximizing potential positive risks.
- The term known risks is sometimes used to describe risks that the project team has identified and analyzed.
- Known risks can be managed proactively.
- However, unknown risks, or risks that have not been identified and analyzed, cannot be managed.
- As you can imagine, good project managers know it is good practice to take the time to identify and manage project risks.
- Six major processes are involved in risk management according to the PMBOK Guide:

1. **Planning risk management** involves deciding how to approach and plan risk management activities for the project. By reviewing the project management plan, project charter, stakeholder register, enterprise environmental factors, and organizational process assets, project teams can discuss and analyze risk management activities for their particular projects. The main output of this process is a risk management plan.
2. **Identifying risks** involves determining which risks are likely to affect a project and documenting the characteristics of each. The main output of this process is the start of a risk register.
3. **Performing qualitative risk analysis** involves prioritizing risks based on their probability of occurrence and impact. After identifying risks, project teams can use various tools and techniques to rank risks and update information in the risk register. The main outputs are project documents updates.

4. **Performing quantitative risk analysis** involves numerically estimating the effects of risks on project objectives. The main outputs of this process are project documents updates.
5. **Planning risk responses** involves taking steps to enhance opportunities and reduce threats to meeting project objectives. Using outputs from the preceding risk management processes, project teams can develop risk response strategies that often result in updates to the project management plan and other project documents.
6. **Controlling risk** involves monitoring identified and residual risks, identifying new risks, carrying out risk response plans, and evaluating the effectiveness of risk strategies throughout the life of the project. The main outputs of this process include work performance information, change requests, and updates to the project management plan, other project documents, and organizational process assets.

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Risk Management Processes by the US DoD Risk Management Guide



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Continuous Risk Management by NASA



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Risk Factors in Software Development

by Steve Naidamast

<https://jaxenter.com/common-sense-software-engineering-part-iii-risk-analysis-121131.html>

Risk type: **Schedule creation**

- Schedule, resources, and project definition have all been dictated by the customer or upper management and are not in balance
- Schedule is optimistic, “best case” (rather than realistic, “expected case”)
- Schedule omits necessary tasks
- Schedule was based on the use of specific team members, but those team members were not available
- Cannot build a project of the size specified in the time allocated
- “Product” of the project is larger than estimated (in lines of code, function points, use case points, or percentage of a previous project’s size)
- Effort is greater than estimated
- Re-estimation in response to schedule slips is overly optimistic or ignores project history
- Excessive schedule pressure reduces productivity
- Target date is moved up with no corresponding adjustments to the project scope or available resources
- A delay in one task causes cascading delays in dependent tasks
- Unfamiliar areas of the product take more time than expected to design and implement

Risk type: **Organisation and management**

- Project lacks an effective top-management sponsor
- Project is too long
- Layoffs and cutbacks reduce team's capacity
- Management or marketing insists on technical decisions that lengthen the schedule
- Inefficient team structure reduces productivity
- Management review/decision cycle is slower than expected
- Budget cuts upset project plans
- Management makes decisions that reduce the project team's motivation
- Nontechnical third-party tasks take longer than expected (budget approval, equipment purchase approval, legal reviews, security clearances etc)
- Planning is too poor to support the desired project speed
- Project plans are abandoned under pressure, resulting in chaotic, inefficient development
- Management places more emphasis on heroics than accurate status reporting, which undercuts its ability to detect and correct problems

Risk type: **Development environment**

- Facilities are not available on time
- Facilities are available but inadequate (e.g. no phones, networking wiring, furniture, office supplies etc)
- Facilities are crowded, noisy or disruptive
- Development tools are not in place by the desired time
- Development tools do not work as expected; developers need time to create workarounds or to switch to new tools
- Development tools are not chosen based on their technical merits and do not provide the planned productivity
- Learning curve for new development tool is longer or steeper than expected

Risk type: **End-users**

- End-user insists on new requirements
- End-user ultimately finds product to be unsatisfactory, requiring redesign and rework
- End-user does not buy into the project and consequently does not provide the needed support
- End-user input is not solicited, so product ultimately fails to meet user expectations and must be reworked

Risk type: **Customer**

- Customer insists on new requirements
- Customer review/decision cycle for plans, prototypes and specifications are slower than expected
- Customer will not participate in review cycles for plans, prototypes and specifications or is incapable of doing so, resulting in unstable requirements and time-consuming changes
- Customer communication time (e.g., time to answer requirements-clarification questions) is slower than expected
- Customer insists on technical decisions that lengthen the schedule.
- Customer micro-manages the development process, resulting in slower progress than planned
- Customer-furnished components are a poor match for the product under development, resulting in extra design and integration work
- Customer-furnished components are poor quality, resulting in extra testing, design and integration work and in extra customer-relationship management
- Customer-mandated support tools and environments are incompatible, have poor performance or have inadequate functionality, resulting in reduced productivity
- Customer will not accept the software as delivered even though it meets all specifications
- Customer has expectations for development speed that developers cannot meet

Risk type: **Contractors**

- Contractor does not deliver components when promised
- Contractor delivers components of unacceptably low quality, and time must be added to improve quality
- Contractor does not buy into the project and consequently does not provide the level of performance needed

Risk type: **Requirements**

- Requirements have been baselined but continue to change
- Requirements are poorly defined and further definition expands the scope of the project
- Additional requirements are added
- Vaguely specified areas of the product are more time-consuming than expected

Risk type: **Product (1)**

- Error-prone modules require more testing, design and implementation work than expected
- Unacceptably low quality and requires more testing, design and implementation work to correct than expected
- Pushing the computer science state-of-the-art in one or more areas lengthens the schedule unpredictably
- Development of the wrong software functions requires redesign and implementation
- Development of the wrong software interface results in redesign and implementation
- Development of extra software functions that are not required (gold-plating) extends the schedule
- Meeting product's size or speed constraints requires more time than expected, including time for redesign and reimplementation

Risk type: **Product (2)**

- Strict requirements for compatibility with existing system required more testing, design and implementation than expected
- Requirements for interfacing with other systems, other complex systems, or other systems that are not under the team's control result in unforeseen design, implementation and testing
- Requirements to operate under multiple operating systems takes longer to satisfy than expected
- Operation in an unfamiliar or unproved software environment causes unforeseen problems
- Operation in an unfamiliar or unproved hardware environment causes unforeseen problems
- Development of a kind of component that is brand new to the organization takes longer than expected
- Dependency on a technology that is still under development lengthens the schedule

Risk type: **External environment**

- Product depends on government regulations, which change unexpectedly
- Product depends on draft technical standards, which change unexpectedly

Risk type: **Personnel (1)**

- Hiring takes longer than expected
- Task prerequisites (e.g. training, completion of other projects, acquisition of work permit) cannot be completed on time
- Poor relationships between developers and management slow decision making and follow through
- Team members do not buy into the project and consequently do not provide the level of performance needed
- Low motivation and more reduced productivity
- Lack of needed specialisation increases defects and rework
- Personnel need extra time to learn unfamiliar software tools or environment
- Personnel need extra time to learn unfamiliar hardware environment
- Personnel need extra time to learn unfamiliar programming language
- Contract personnel leave before project is complete
- Permanent employees leave before project is complete
- New development personnel are added late in the project, and additional training and communications overhead reduces existing team members' effectiveness
- Team members do not work together efficiently

Risk type: **Personnel (2)**

- Conflicts between team members result in poor communication, poor designs, interface errors and extra rework
- Problem team members are not removed from the team, damaging overall team motivation
- The personnel most qualified to work on the project are not available for the project
- The personnel most qualified to work on the project are available for the project but are not used for political or other reasons
- Personnel with critical skills needed for the project cannot be found
- Key personnel are available only part time
- Not enough personnel are available for the project
- People's assignments do not match their strengths
- Personnel work slower than expected
- Sabotage by project management results in inefficient scheduling and ineffective planning
- Sabotage by technical personnel results in lost work or poor quality and requires rework

Risk type: **Design and implementation**

- Overly simple design fails to address major issues and leads to redesign and reimplementation
- Overly complicated design requires unnecessary and unproductive implementation overhead
- Poor design leads to redesign and reimplementation
- Use of unfamiliar methodology results in extra training time and rework to fix first time misuses of the methodology
- Product is implemented in a low-level language and productivity is lower than expected
- Necessary functionality cannot be implemented using the selected code or class libraries; developers must switch to new libraries or custom-build the necessary functionality
- Code or class libraries have poor quality, causing extra testing, defect correction and rework
- Schedule savings from productivity enhancing tools are overestimated
- Components developed separately cannot be integrated easily, requiring redesign and rework

Risk type: **Process**

- Amount of paperwork results in slower progress than expected
- Inaccurate progress tracking results in not knowing the project is behind schedule until late in the project
- Upstream quality assurance activities are shortchanged, resulting in time-consuming rework downstream
- Inaccurate quality tracking results in not knowing about quality problems that affect the schedule until late in the project
- Too little formality (lack of adherence to software policies and standards) results in miscommunications, quality problems and rework
- Too much formality (bureaucratic adherence to software policies and standards) results in unnecessary, time-consuming overhead
- Management-level progress reporting takes more developer time than expected
- Half-hearted risk management fails to detect major project risks
- Software project risk management takes more time than expected



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Top 10 Software Development Risks

<https://www.itproportal.com/2010/06/14/top-ten-software-development-risks/>

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1. Estimation and scheduling

- The unique nature of individual software projects creates problems for developers and managers in estimating and scheduling development time.
- Always monitor existing projects so that you apply lessons learnt in the future.

2. Sudden growth in requirements

- As the project progresses, unidentified issues can make it difficult to meet deadlines at the last minute.
- Try to think boldly at the beginning of the project and anticipate the worst or most intense scenario.

3. Employee turnover

- Every project has a number of developers working on it.
- When a developer leaves, he or she may take critical information with him/her.
- This can delay, and sometimes derail an entire project.
- Ensure you have resources where team members can collaborate and share knowledge.

Top 10 Software Development Risks

<https://www.itproportal.com/2010/06/14/top-ten-software-development-risks/>

4. Breakdown of specification

- During the initial phases of integration and coding, requirements might conflict.
- Moreover, developers may find that even the specification is unclear or incomplete.

5. Productivity issues

- On projects involving long timelines, developers tend to take things easy to begin with.
- As a result, sometimes, they lose significant time to complete the project.
- Set a realistic schedule, and stick to it.

6. Compromising on designs

- In order to get stuck into the next ‘real’ tasks, developers tend to rush the design-process.
- This is a waste of programming hours, as designing is the most critical part of software development.

7. Gold plating

- Developers sometimes like to show off their skills by adding unnecessary features. For instance, a developer might add Flash to a basic login module to make it look ‘stylish’. Again, this is a waste of programming hours.

Top 10 Software Development Risks

<https://www.itproportal.com/2010/06/14/top-ten-software-development-risks/>

8. Procedural risks

Day-to-day operational activities might hamper due to improper process implementation, conflicting priorities, or a lack of clarity in responsibilities.

9. Technical risks

- Sometimes software development firms reduce the functionality of the software to compensate for overruns pertaining to high budgets and scheduling.
- There is always a conflict between achieving maximum functionality of the software and peak performance.
- In order to compensate for excessive budget and schedule overruns, companies sometimes reduce the functionality of the software.

10. Unavoidable risks

- These include changes in government policy, the obsolescence of software or other risks that cannot be controlled or estimated.
- As the field of software development becomes more and more complex, the risks associated with it have intensified.
- It is vital that development firms focus on strategic planning to mitigate such risks.



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Risk Matrix



Risk Matrix

- A risk matrix is a matrix that is used during qualitative risk assessment to define the level of risk by considering the category of probability (or likelihood) against the category of consequence (impact) severity.
- This is a simple mechanism to increase visibility of risks, prioritize risks, and assist management decision making.

Risk Assessment Matrix				
Impact of Risk (Consequence)	Major	Medium	High	Extreme
	Moderate	Medium	Medium	High
	Minor	Low	Medium	Medium
Seriousness of Risk = Probability x Impact		Unlikely (0-33%)	Moderately Likely (33%-66%)	Highly Likely (66%-100%)
		Probability of Risk (Likelihood)		

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	SEVERITY →		
	1	2	3
LIKELIHOOD ↓	1	LOW 1	MEDIUM 3
	2	LOW 2	HIGH 6
	3	MEDIUM 3	HIGH 9

This 3x3 risk matrix template is ideal for teams and organizations that prefer simplicity.

The template provides three levels to code both the severity and likelihood of each risk: low, medium, and high (which are assigned values of one, two, and three, respectively).

After determining the values for the severity and likelihood, use the grid to determine the risks that need to be addressed first.

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		SEVERITY →			
		ACCEPTABLE LITTLE TO NO EFFECT ON EVENT	TOLERABLE EFFECTS ARE FELT, BUT NOT CRITICAL TO OUTCOME	UNDESIRABLE SERIOUS IMPACT TO THE COURSE OF ACTION AND OUTCOME	INTOLERABLE COULD RESULT IN DISASTER
LIKELIHOOD ↓	IMPROBABLE RISK IS UNLIKELY TO OCCUR	LOW 1	MEDIUM 4	MEDIUM 6	HIGH 10
	POSSIBLE RISK WILL LIKELY OCCUR	LOW 2	MEDIUM 5	HIGH 8	EXTREME 11
	PROBABLE RISK WILL OCCUR	MEDIUM 3	HIGH 7	HIGH 9	EXTREME 12

This 3x4 risk matrix template uses non-numeric scales for likelihood and severity; after selecting the options for each parameter, use the values in the matrix to determine the level of severity for each risk.

RISK RATING KEY	LOW 0 – ACCEPTABLE	MEDIUM 1 – ALARP as low as reasonably practicable	HIGH 2 – GENERALLY UNACCEPTABLE	EXTREME 3 – INTOLERABLE
	OK TO PROCEED	TAKE MITIGATION EFFORTS	SEEK SUPPORT	PLACE EVENT ON HOLD

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		SEVERITY →				
		1	2	3	4	5
LIKELIHOOD ↓	1	LOW 1	LOW 2	LOW 3	MEDIUM 4	MEDIUM 5
	2	LOW 2	MEDIUM 4	MEDIUM 6	HIGH 8	HIGH 10
	3	LOW 3	MEDIUM 6	HIGH 9	HIGH 12	EXTREME 15
	4	MEDIUM 4	HIGH 8	HIGH 12	HIGH 16	EXTREME 20
	5	MEDIUM 5	HIGH 10	EXTREME 15	EXTREME 20	EXTREME 25

The 5x5 matrix template provides additional scaling options; this template is ideal for organizations that need more granular insight into each risk.

After determining the values for severity and likelihood, use the grid to determine the priority of the risks.

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		Consequence					
		How severe could the outcomes be if the risk event occurred? →					
		1 Insignificant	2 Minor	3 Significant	4 Major	5 Severe	
Likelihood	What's the chance the risk occurring? ↑	5 Almost Certain	5 Medium	10 High	15 Very high	20 Extreme	25 Extreme
		4 Likely	4 Medium	8 Medium	12 High	16 Very high	20 Extreme
		3 Moderate	3 Low	6 Medium	9 Medium	12 High	15 Very high
		2 Unlikely	2 Very low	4 Low	6 Medium	8 Medium	10 High
		1 Rare	1 Very low	2 Very low	3 Low	4 Medium	5 Medium

Risk Matrix

Rating likelihood of risks:

- **Unlikely**: Extremely rare risks, with almost no probability of occurring.
- **Seldom**: Risks that are relatively uncommon, but have a small chance of manifesting.
- **Occasional**: Risks that are more typical, with about a 50/50 chance of taking place.
- **Likely**: Risks that are highly likely to occur.
- **Definite**: Risks that are almost certain to manifest.

Risk Matrix

Rating impact of risk:

- **Insignificant:** Risks that bring no real negative consequences, or pose no significant threat to the organization or project.
- **Minor:** Risks that have a small potential for negative consequences, but will not significantly impact overall success.
- **Moderate:** Risks that could potentially bring negative consequences, posing a moderate threat to the project or organization.
- **Critical:** Risks with substantial negative consequences that will seriously impact the success of the organization or project.
- **Catastrophic:** Risks with extreme negative consequences that could cause the entire project to fail or severely impact daily operations of the organization. These are the highest-priority risks to address.

Classifying and Prioritizing Risks

- After you've placed each risk in the matrix, you can give it an overall risk ranking.
- Risks that have severe negative consequences and are highly likely to occur receive the highest rank; risks with both low impact and low likelihood receive the lowest rank.
- Risk rankings combine impact and likelihood ratings to help you identify which risks pose the greatest overall threats (and therefore are the top priority to address).
- Some organizations use a numeric scale to assign more specific risk rankings.
- However, most rankings fall into a few broad categories, which are often color-coded:

Classifying and Prioritizing Risks

- **Low:** The consequences of the risk are minor, and it is unlikely to occur. These types of risks are generally ignored, and often color-coded green.
- **Medium:** Somewhat likely to occur, these risks come with slightly more serious consequences. If possible, take steps to prevent medium risks from occurring, but remember that they are not high-priority and should not significantly affect organization or project success. These risks are often color-coded yellow.
- **High:** These are serious risks that both have significant consequences, and are likely to occur. Prioritize and respond to these risks in the near term. They are often color-coded orange.
- **Extreme:** Catastrophic risks that have severe consequences and are highly likely to occur. Extreme risks are the highest priority. You should respond to them immediately, as they can threaten the success of the organization or project. They are often color-coded red.

The advantages of a risk matrix

A risk matrix helps to:

- Prioritize risks according to their level of severity;
- In risk planning, it helps to neutralize possible consequences;
- Allows the analysis of potential risks with minimum effort;
- It improves the security measures of the organization;
- Gives the team an overview of the potential risks of a project.

Note: A risk matrix is a tool and not a complete solution for the needs of a project.



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Risk Register

Risk Register

- The main output of risk identification is a list of identified risks and other information needed to begin creating a risk register.
- A risk register is a document that contains results of various risk management processes; it is often displayed in a table or spreadsheet format.
- A risk register is a tool for documenting potential risk events and related information.
- Risk events refer to specific, uncertain events that may occur to the detriment or enhancement of the project.
- Risk Register is the most important document in the project management.
- Risk Register is a document that contains the information about identified risks, results of Risk Analysis (impact, probability, effects), as well as Risk Response Plans.
- You also use the Risk Register to monitor and control risks during the whole project life cycle.

Risk Register by PMBOK Guide

- The risk register captures details of identified individual project risks.
- The risk register may contain limited or extensive risk information depending on project variables such as size and complexity.
- On completion of the Identify Risks process, the content of the risk register may include but is not limited to:
- **List of identified risks.** Each individual project risk is given a unique identifier in the risk register. Identified risks are described in as much detail as required to ensure unambiguous understanding. A structured risk statement may be used to distinguish risks from their cause(s) and their effect(s).
- **Potential risk owners.** Where a potential risk owner has been identified during the Identify Risks process, the risk owner is recorded in the risk register. This will be confirmed during the Perform Qualitative Risk Analysis process.
- **List of potential risk responses.** Where a potential risk response has been identified during the Identify Risks process, it is recorded in the risk register. This will be confirmed during the Plan Risk Responses process.

Risk Register by PMBOK Guide

- Additional data may be recorded for each identified risk, depending on the risk register format specified in the risk management plan.
- This may include:
 - a short risk title,
 - risk category,
 - current risk status,
 - one or more causes,
 - one or more effects on objectives,
 - risk triggers (events or conditions that indicate that a risk is about to occur),
 - WBS reference of affected activities,
 - and timing information
 - when was the risk identified,
 - when might the risk occur,
 - when might it no longer be relevant,
 - and what is the deadline for taking action.

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Example of Risk Register



Risk Register

Project name: Common project risks

Project manager: J Black

ID	Date raised	Risk description	Likelihood of the risk occurring	Impact if the risk occurs	Severity <i>Rating based on impact & likelihood.</i>	Owner <i>Person who will manage the risk.</i>	Mitigating action <i>Actions to mitigate the risk e.g. reduce the likelihood.</i>	Contingent action <i>Action to be taken if the risk happens.</i>	Progress on actions	Status	Useful resources
1	[enter date]	Project purpose and need is not well-defined.	Medium	High	High	Project Sponsor	Complete a business case if not already provided and ensure purpose is well defined on Project Charter and PID.	Escalate to the Project Board with an assessment of the risk of runaway costs/never-ending project.	Business case re-written with clear deliverables and submitted to the project board for approval.	Open	Business Case Template
2	[enter date]	Project design and deliverable definition is incomplete.	Low	High	High	Project Sponsor	Define the scope in detail via design workshops with input from subject matter experts.	Document assumptions made and associated risks. Request high risk items that are ill-defined are removed from scope.	Design workshops scheduled.	Open	Assumptions Log
3	[enter date]	Project schedule is not clearly defined or understood	Low	Medium	Medium	Project Manager	Hold scheduling workshops with the project team so they understand the plan and likelihood of missed tasks is reduced.	Revisit the schedule with the project team. 'Relaunch' the schedule.	Workshops scheduled.	Open	Project Status Template
4	[enter date]	No control over staff priorities	Medium	Medium	Medium	Project Manager	The Project Sponsor will brief team managers on the importance of the project. Soft book resources as early as possible and then communicate final booking dates asap after the scheduling workshops. Identify back ups for each human resource on the project.	Escalate to the Project Sponsor and bring in back up resource.	Project Sponsor has agreed to hold briefing. Now making arrangements for a meeting room.	Open	Who is the Project Sponsor?
5	[enter date]	Consultant or contractor delays	Medium	High	High	Project Manager	Include late penalties in contracts. Build in and protect lead time in the schedule. Communicate schedule early. Check in with suppliers regularly. Query '90% done'. Ask again and again if they need anything else.	Escalate to Project Sponsor and Contracts Manager. Implement late clauses.	Lead time from each contractor built into the project schedule. Late penalties agreed to and contracts signed.	Open	What is lead time?

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Example of Risk Register

Project Risk Register

Edge Lane

Project

mouchel

Davis Langdon

2020

Liverpool

LIVERPOOL

VISION

The City of Liverpool

Project Title	Edge Lane West Highway and Environmental Improvement Scheme					Mouchel Project Manager		Mouchel Project Number		Last Updated				
						Chris Founds		760155		May-09				

2.0 Design Risks

Risk ID	Entry by	Risk (description)	Date Raised	Impact	Probability	Project Severity	Mitigation measures	Ownership	Adjusted Impact	Adjusted Probability	Adjusted Severity	Status	Last Review Date	Date closed
2.1		Utilities and Services												
2.1.1	Mou	Inaccurate information affects proposed design and layout	July05	H	M	H	Radar surveys carried out to specified lengths where planting envisaged. Trial trenches undertaken and all service locations accurately measured. Revisit service details with SAs before and at ECI Stage 1 and engage early in process. Further trial pitting to be undertaken by Contractor prior to start on site to confirm	Mouchel	H	L	M	Radar surveys complete. Revised C4 Estimates being obtained	June '09	
2.1.2	Mou	Not being informed of Stats own strategic and investment / maintenance and development programmesthat could impact on adjacent routes	July05	M	H	H	Regular dialogue with Stat Authorities through co-ordination meetings with LCC. Any proposed works flagged up at this meeting to be checked with relevant authority.	Mouchel, LCC	M	M	M	In hand but worth noting that not all proposed diversions/works are known to attendees at co-ordination meeting.	June '09	
2.1.3	Mou	Additional Utilities provision required for adjacent development plots that are nor known about and affect Utilities proposals and resources	July05	H	H	H	Co-ordination to be discussed through Council Development Sub-Group and requirements understood	Vision	H	L	M	Bellway and PCT furthest developed in terms of Utilities requirements	June '09	
2.1.4	Mou	Lack of co-operation response due to current level of activity around Liverpool	July05	H	M	H	Close liaison and planning well in advance	Mouchel	M	L	L	Ongoing	June '09	
2.2		Design Management												
2.2.1	Mou	Failure to meet agreed programme milestones for design delivery before and during ECI period	July05	H	L	M	Internal design team meetings identifying potential problems well in advance to ensure ability for solutions and production of timely information.	Mouchel	M	L	L	Previous experience of 4 ECI projects along Edge Lane has established sound processes	June '09	
2.2.2	Mou	Failure to agree responsibility for interface boundaries with developers	July05	L	M	L	Detailed dialogue between Developers and Mouchel have been productive and will continue through Development Sub Group meetings	Mouchel, Vision	L	L	L	Responsibilities now defined for PCT and Bellway sites. Some site procedures still to detail up. Lines of communication to remain open	June '09	
2.2.3	Mou	Buildability of proposed details	July05	H	L	M	ECI form of contract selected. Discussions with selected contracrtactors to take place well in advance of final construction details being produced	Mouchel	L	L	L	Build on previous work undertaken by original Framework Contractors and use lessons learnt and best practice from previous phases of Edge Lane	June '09	
2.2.4	Mou	S278 Design approval by LCC throws up significant comments and issues	July05	H	L	M	Close liaison with LCC to ensure their acceptance of design and materials. Proposals similar to previously approved and completed sections of Edge Lane	Mouchel / 2020	M	L	L	Ongoing discussions. First package due to be sent to LCC w/c 25th July for technical approval checking/approval.	June '09	



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Risk Response Strategies

5 Main Risk Response Strategies (for Threats)

<https://pmbasics101.com/really-best-risk-response-strategy/>

Examples of Valid Negative Risks (Threats) Responses

1. Avoid
2. Mitigate
3. Transfer
4. Actively Accept
5. Passively Accept

5 Main Risk Response Strategies (for Threats)

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1. Avoid

It means you need to do something to eliminate the cause of the threat, for example:

- Remove a work package or delivery from WBS to secure delivery of the rest of the project.
- Remove a conflicting team member to stop demotivation in the team.
- Forbid any work in bad weather to avoid the risk that someone will get hurt.

5 Main Risk Response Strategies (for Threats)

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2. Mitigate

Do something to reduce the impact or the probability of a threat, for example :

- Prototype unclear or risk delivery early on to get early feedback from a customer.
- Plan frequent visits to a vendor to learn about problems as early as possible.
- Train the team in risk management approach.

5 Main Risk Response Strategies (for Threats)

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3. Transfer

Take action to make another party responsible for the risk, for example:

- Outsource part of a project.
- Buy insurance on the property.
- Employ a part-time legal or procurement expert.

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4. Actively Accept

It means that you need to develop a (contingency) plan and make reserves for a risk. However, you will only act if and when the risk happens. For example:

- If a critical person gets sick – we will get a substitution.
- If a work package takes more time, we will work overtime.
- If the equipment breaks, we will buy a new one using reserves.

5. Passively Accept

- Do really nothing. If a risk happens, you will need to decide if there is a workaround.

5 Main Risk Response Strategies (for Opportunities)

<https://pmbasics101.com/really-best-risk-response-strategy/>

Examples of Positive Risk Response Strategies (Opportunities)

1. Exploit
2. Enhance
3. Share
4. Actively Accept
5. Passively Accept

5 Main Risk Response Strategies (for Opportunities)

<https://pmbasics101.com/really-best-risk-response-strategy/>

1. Exploit

Do some extra work or change the project plan to make an opportunity happen, for example :

- Plan risky work packages for the most experienced team members.
- Suggest a better approach to reduce the required efforts.
- Suggest a solution to get a new contract from the client.
- Finish the current project earlier to get another project.

5 Main Risk Response Strategies (for Opportunities)

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2. Enhance

Do something to increase the chances or impact of an opportunity, for example :

- Buy the equipment beforehand when the price is lower.
- Negotiate the transfer of exceptional expert to your team as early as possible.
- Promise incentives to the team to finish a project beforehand to start a new one.

5 Main Risk Response Strategies (for Opportunities)

<https://pmbasics101.com/really-best-risk-response-strategy/>

3. Share

Share benefits with another party for an opportunity to happen for both of you, for example :

- Create a partnership with a third party to achieve your goals.

4. , 5. Actively and Passively Accept

- You can Actively and Passively Accept opportunities as well as threats.

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Risk Register Template

- Risk Register is a document that contains the information about identified risks, results of Risk Analysis (impact, probability, effects), as well as Risk Response Plans.
- You also use the Risk Register to monitor and control risks during the whole project life cycle.
- Risk Register should be simple, adaptable, maintainable, and close at hand.

Project Name	My Project
Last Revision Date	12.12.2017

Index	WBS Element	Category	Description	Effects	Probability	Impact	Risk Rank	Owner	Response Plan
0001	1.1.3	Vendors	Web component of the application depends on the design created by the external design team.	Delays in the design delivery will directly impact the project schedule	6	10	60	D. Nizhebetkiy	"Procurement manager should include the following terms in the contract: - Weekly Reports. - Weekly on-site meetings. - Delivery via collaboration system (e.g. Invision)"
0002	1.6.5	HR	Resources for mobile development are limited and on	- Unavailability of developers may cause delays. - Quality may suffer due to multitasking.	8	8	64	D. Nizhebetkiy	- Develop cross-project HR plan together with Ann Smith and Ron Nagle - Secure required resources.
0003	1.2.3.6	Quality	Implementation of service may require changes to the infrastructure.	- Poor performance. - Delivery will not be accepted.	3	10	30	R. Anderson	- Reserve 2 days of work in the beginning of January for Infrastructure Team. - Perform regular performance testing. - Reserve time for performance optimisation (about 2 weeks).
0004	1,3	Scope	Implementation depends on delivery 1.1.3.	- More efforts will be required to implement the UI.	3	7	21	J. Robertson	- BSA should control the vendor on regular basis - Final designs should not require any unsupported technologies.
0005	1.3.2.1	HR	Only one engineer in the company has required skills to deliver the work package. He is currently in conflict with the functional manager. There is a good chance that he will leave the company.	- Delivery will be fully blocked.	6	10	60	A. Lee E. Remington	- Request HR department to open new vacancy. - Request HR to find an outsourcing/freelancing options. - Help to resolve conflict asap.

Checklist of best practices for managing risk

A checklist of best practices for managing risk on software development and software engineering projects is as follows:

- Always be forward-thinking about risk management. Otherwise, the project team will be driven from one crisis to the next.
- Use checklists, and compare with similar previous projects.
- Prioritize risks, ranking each according to the severity of exposure.
- Develop a top-10 or top-20 risk list for your project. Like most project managers, you can probably reuse this list on the next project!
- Vigorously watch for surfacing risks by meeting with key stakeholders—especially with the marketing team and the customer.
- As practicable, split larger risks into smaller, easily recognizable and readily-manageable risks.
- Strongly encourage stakeholders to think proactively and communicate about risks throughout the entire project.



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Thank You very much
for your attention